Steam & Utilities System SOP — Bioethanol Plant (excerpt)

Supplied grounding document for capstone groups (e.g. a utilities/steam agent) that don't have their own. Covers the LP steam system feeding reboiler E-201 and general plant utilities.

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1. Purpose & scope

Defines normal operation, alarms, and safe limits for the low-pressure (LP) steam distribution system, boiler B-101, and general plant utilities (instrument air, cooling water supply header, electrical). Applies to utilities operators.

2. Steam system normal parameters

Parameter	Normal	Safe range	Trip
Boiler B-101 steam header pressure	6.0 bar g	5.5–6.5 bar g	7.5 bar g → safety relief
LP distribution pressure (to E-201)	3.5 bar g	3.0–4.0 bar g	—
Boiler drum level	Normal (50%) band	30–70%	< 20% → low-level trip / burner cutout
Feedwater temperature to boiler	≥ 90 °C	85–105 °C	—
Condensate return temperature	≤ 95 °C	—	—

D-201's reboiler steam demand (1.8–3.0 t/h normal, SOP-D201-014 §3) is the largest single draw on the LP header; a distribution pressure drop during high-demand periods elsewhere on site can show up first as a struggling reboiler duty on D-201.

3. Instrument air & general utilities

- Instrument air header: minimum 6 bar g required for control valves and trip actuation; low instrument air pressure can cause control valves to fail to their safe position (fail-closed/fail-open per valve design) — this is a safety function, not a fault to bypass.
- Electrical supply: utilities-critical loads (boiler feedwater pumps, instrument air compressors) are on the essential/backup supply; a loss of essential power is an immediate escalation, not a troubleshooting item.

4. Troubleshooting (quick guide)

Symptom	Likely cause	First checks
LP distribution pressure sagging under load	Boiler capacity limit, or a large simultaneous demand elsewhere	Check B-101 firing rate and header pressure trend; do not rob pressure by closing other users' valves without utilities-engineer approval
Reboiler E-201 struggling to hold duty	LP pressure low, or steam trap/valve fault on the E-	Confirm LP header pressure first; if header is

	201 supply	normal, suspect a local trap/valve issue
Boiler drum level trending low	Feedwater pump issue, or high steam demand outstripping feed	Check feedwater pump status and W-301 water quality (see Water Treatment SOP) before adjusting boiler firing

5. Escalation & human sign-off

- The boiler safety relief, drum-level trip, and instrument-air fail-safe actions are safety functions and must never be bypassed or reset by an operator or assistant — only a qualified utilities engineer, under permit where required.
- An assistant may advise on likely causes of a steam or utility upset and cite this SOP, but must not recommend changing boiler firing rate, header pressure setpoints, or bypassing any trip; these require utilities-engineer sign-off.